

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459623.9004 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.016
Transit depth (Rp/Rs)^2	2.56 +/- 0.5 [%]
Orbital Inclination (inc)	86.58 +/- 1.39
Airmass coefficient 1 (a1)	1.212 +/- 0.021
Airmass coefficient 2 (a2)	-0.128 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.71 %
Best Comparison Star	None
Optimal Aperture	3.82
Optimal Annulus	11.89
Transit Duration (day)	0.0599 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.016 vs 0.1594 (deviation 0.03σ)
Duration fit vs published	0.0599 d vs 0.0512 d (deviation 3.96σ)
Mid-time O–C	4.7 min
Depth SNR	8.2
Residual scatter	1.71 %

Light curve

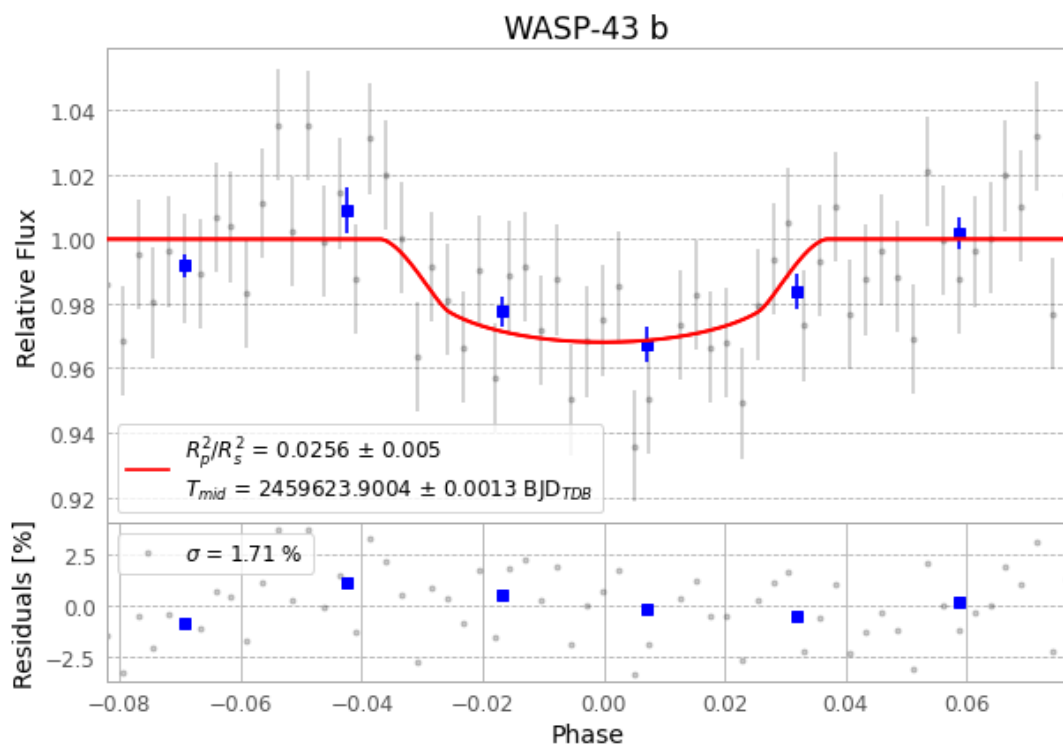


Figure 1 — FinalLightCurve_WASP-43 b_2022-02-13.png (SHA-256 b82b1bb958a5483c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [312, 247], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9f3cae5d3eb89a8e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), WASP-43220213075112.FITS → WASP-43220213105712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220213.log (5d07cc31b524...), FinalLightCurve_WASP-43 b_2022-02-13.png (b82b1bb958a5...), FinalLightCurve_WASP-43 b_2022-02-13.csv (b171d5a5de0e...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 20:12Z.